

Ana Gainaru

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PROFESSIONAL EXPERIENCE	Computer Scientist <i>Oak Ridge National Laboratory, CSMD</i>	Sep 2020 - present
	Research Assistant Professor <i>Vanderbilt University, Electrical Engineering and Computer Science Department</i>	Jul 2017 - Aug 2020
	HPC Architect <i>Mellanox Technologies Inc</i>	Oct 2015 - Jun 2017
EDUCATION	University of Illinois at Urbana-Champaign, IL, USA <i>PhD in Computer Science</i>	Aug 2011 - Aug 2015
	University Politehnica of Bucharest, Romania <i>Master of Science in Computer Science</i>	Oct 2008 - Jul 2010
	University Politehnica of Bucharest, Romania <i>Bachelor of Science in Computer Science</i>	Oct 2003 - Jul 2008
RELEVANT PUBLICATIONS	1. Greg Eisenhauer, Norbert Podhorszki, Ana Gainaru, Scott Klasky, Junmin Gu, Vicente Bolea, et al. <i>HPC I/O innovations in the exascale era</i> [The International Journal of High Performance Computing Applications, 2025] 2. Wes Brewer, Ana Gainaru, Frédéric Suter, Feiyi Wang, Murali Emani, Shantenu Jha <i>AI-coupled HPC Workflow Applications, Middleware and Performance</i> [ACM Computing Surveys Journal (CSUR), 2025] 3. Norbert Podhorszki, Greg Eisenhauer, Tahsin Kurc, Qian Gong, Qing Liu, Jieyang Chen, Ana Gainaru, Junmin Gu, Scott Klasky <i>HPC Campaign Management: Remote data access with user-defined error bound using ADIOS and ZFP</i> [Supercomputing Asia Conference, 2025] 4. Jaime Cernuda, Luke Logan, Ana Gainaru, Scott Klasky, Jay Lofstead, Anthony Kougkas, Xian-He Sun. <i>Hades: A Context-Aware Active Storage Framework for Accelerating Large-Scale Data Analysis</i> [IEEE 24th International Symposium on Cluster, Cloud and Internet Computing (CCGrid), 2024] 5. A. Gainaru, N. Podhorszki, L. Dulac, Q. Gong, S. Klasky, G. Eisenhauer, A. Kougkas, X. Sun, J. Lofstead <i>To Derive or Not to Derive: I/O Libraries Take Charge of Derived Quantities Computation</i> [IEEE 36th International Symposium on Computer Architecture and High Performance Computing (SBAC-PAD), 2024] 6. Lipeng Wan, Jieyang Chen, Xin Liang, Ana Gainaru, Qian Gong, Qing Liu, Ben Whitney, et al. <i>RAPIDS: Reconciling Availability, Accuracy, and Performance in Managing Geo-Distributed Scientific Data</i> [HPDC: Proceedings of the 32nd International Symposium on High-Performance Parallel and Distributed Computing, 2023] 7. Frédéric Suter, Rafael Ferreira da Silva, Ana Gainaru, Scott Klasky. <i>Driving Next-Generation Workflows from the Data Plane</i> [e-Science 2023: 1-10] 8. Eric Suchyta, Jong Youl Choi, Seung-Hoe Ku, David Pugmire, Ana Gainaru, Kevin A. Huck, Ralph Kube, Aaron Scheinberg, Frédéric Suter, Choong-Seock Chang, Todd S. Munson, Norbert Podhorszki, Scott Klasky. <i>Hybrid Analysis of Fusion Data for Online Understanding of Complex Science on Extreme Scale Computers</i> [CLUSTER 2022: 218-229] 9. Ana Gainaru, Lipeng Wan, Ruonan Wang, Eric Suchyta, Jieyang Chen, Norbert Podhorszki, James Kress, David Pugmire, Scott Klasky: <i>Understanding the Impact of Data Staging for Coupled Scientific Workflows</i>. [Parallel Distributed Syst. 33(10): 4134-4147 (2022)]	

10. Mark Cianciosa, Richard Archibald, Wael Elwasif, [Ana Gainaru](#), Jin Myung Park, Ross Whitfield *Adaptive Generation of Training Data for ML Reduced Model Creation* [2022 IEEE International Conference on Big Data : 3408-3416]

INVITED TALKS

1. *On demand scheduling for large-scale workflows with analysis requirements* [18th Scheduling for large-scale systems workshop, 2025]
2. *Efficient Digital Twin Training using Uncertainty-Guided Data Generation* [SIAM CSE25 workshop: Neural Acceleration, Surrogate Models, and Learning Techniques for HPC Kernels, 2025]
3. *Schema for the interactions between HPC, AI frameworks and visalization* [Open Data Schema Workshop, 2025]
4. *I/O scheduling for large workflows coupling in near-real time AI training with simulations* [17th Scheduling for Large Scale Systems workshop, 2024]
5. *Benchmarking Large Language Models on Large Scale Systems* [Blueprinting AI for Science Project STFC UKRI, 2024]
6. *A data-centric view on workflows that couple HPC with large-scale models* [Workshop on Advancing Neural Network Training (WANT) at NeurIPS, 2023]
7. *I/O patterns in AI workflows and their impact on the performance of HPC simulations* [Dagstuhl Seminar: Integrating HPC, AI, and Workflows for Scientific Data Analysis, 2023]
8. *How to efficiently abuse your HPC system in the age of AI workflows* [Seminar Series, Science and Technology Facilities Council (STFC), 2023]
9. *Scheduling the I/O of AI applications with a focus on medical imaging* [16th Scheduling for Large Scale Systems workshop, 2023]

PANEL

Challenges to scale model training on HPC systems [Workshop on Advancing Neural Network Training (WANT) at NeurIPS, 2023] Panelists: [Ana Gainaru](#), Lionel Eyraud-Dubois, Tunji Ruwase, Natalia Vassilieva, Mohammad Shoeybi, Jean Kossaifi

Panel on Unspoken Challenges [The 16th Scheduling for Large Scale Systems workshop, 2023] Panelists: George Bosilica, Franck Cappello, [Ana Gainaru](#), Al Geist, Yves Roberts

Data staging and coordination strategies for coupled simulation workflows [ECP 2021]

Moderator: Ana Gainaru,

Panelists: Hank Childs, Dave Pugmire, Greg Eisenhauer, Scott Klasky, Bogdan Nicolae, Pradeep Subedi, Stephane Ethier

Fault Tolerance/Resilience at Petascale/Exascale: Is it Really Critical? Are Solutions Necessarily Disruptive? [SC 2013]

Panelists: Marc Snir, Bronis De Supinski, Al Geist, John Daly, [Ana Gainaru](#) , Satoshi Matsuoka

PATENTS

Richard Graham, [Ana Gainaru](#)

Using Hardware Gather-Scatter Capabilities to Optimize MPI All-to-All

U.S. Patent No. 15/681, 390, 20 August 2017

WORKSHOP ORGANIZING

- The Workshop on Advancing Neural Network Training (WANT): Computational Efficiency, Scalability, and Resource Optimization, 2nd edition, in conjunction with NeurIPSs 2024
- The Workshop on Monitoring, Observability, and in situ Analytics (WOWMON), 1st edition, in conjunction with ICPP 2024
- The International Workshop on Data Analysis and Reduction for Big Scientific Data (DRBSD), 8-11th editions, in conjunction with SC 2022-2025
- The Convergence Computing Infrastructure Workshop (CCIW), 1st and 2nd editions, in conjunction with HPDC 2019-2020

PROFESSIONAL
AND COMMUNITY

- NSF Panel review member, 2019/2020/2021/2025
- ASCR Panel Review member 2021/2022/2023/2024/2025
- Journal editorial board: International Journal for High Performance Computing Applications (IJHPCA), 2019-currently still on the editorial board IEEE Transactions on Parallel and Distributed Systems (TPDS), part of the Review Board for an 18 month term since February 2020; Journal of Parallel and Distributed Computing (JPDC) as as a subject area editor (AE) 2019;
- Vice chair or Co-chair:
 - Vice chair for the HPC Asia 2023, Tutorials SC 2020 Committee, Poster SC 2018 Committee
 - Co-chair for ICPP 2025: Performance track, CCGrid 2024 for Track 2: Software Systems and Platforms, Best ACM SRC Posters at SC 2023, Doctoral showcase at SC 2022, FTS 2017 Workshop on Fault Tolerant Systems in conjunction with CCGRID 2017

I have been on the program committee for over 35 conferences and workshops in the past 10+ years in areas related to HPC and algorithms. Conferences include HPDC, SC paper tracks/BOF/Poster, IPDPS, Cluster, ICPP, ICS, CCGRID, EuroMPI, ISC.